PHYSICS: ELECTROMAGNETISM & SOLID-STATE TRANSPORT

CHAPTER 1: ELECTRIC POTENTIAL & DRIFT DYNAMICS

1.1 Applied Electric Field & Conduction Electrons

When an external electric field is applied across a solid metallic conductor, free conduction electrons experience a steady electrostatic force F = -eE. Rather than accelerating infinitely without bound, mobile charge carriers undergo repetitive non-elastic scattering against the positive ionic crystal lattice. This mechanical drag opposes the electrostatic push, rapidly establishing a steady terminal drift velocity vd.

1.2 Mean Free Path and Relaxation Time

Between successive collisions with crystal defects and vibrating lattice phonons, conduction electrons accelerate freely over an average characteristic timescale known as the momentum relaxation time tau. The steady-state drift velocity is governed by the equilibrium relation vd = (e * tau / m) * E, where m represents the effective electronic mass.

1.3 Electric Potential Difference as an Energy Gradient

The macroscopic electric potential difference Delta V represents the work done per unit charge available to transport mobile electrons across this dense lattice barrier. A steeper potential gradient imparts greater mechanical acceleration between collision events, establishing a stronger driving pressure across the conductor.

1.4 Volumetric Charge Flux and Current Density

Macroscopic electric current I is directly proportional to the number density of charge carriers n, the elementary electron charge e, the conductor cross-sectional area A, and the steady-state drift velocity: I = n * e * A * vd. When charge carriers scatter against the ionic lattice, kinetic energy gained from the potential drop is irreversibly transferred into lattice vibrational modes, generating thermal dissipation.

1.5 Fundamental Conservation Invariant: Current Continuity

Under steady-state conditions, electric charge cannot spontaneously accumulate or deplete at any point within a closed conductor. The net flux of mobile charge entering any closed volume must exactly equal the net charge exiting that volume. Current continuity represents an inviolable topological invariant across all conductive circuits.